WITH R^1 KILLED AND CURVE FIBRES, THE STRUCTURE PUSHFORWARD IS S-FLAT AND COMMUTES WITH BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 21192-21206 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 21208-21232, the complete author proof (23 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Third member of the chapter's h1 family (with L16077 propagation and L16097 detection, both collected): this is the flatness-of-H0 theorem the family builds toward. The two author-flagged detail omissions are both one-line standard bookkeeping, every substantive input is named.

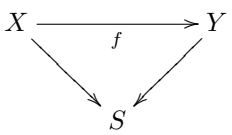
Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The chapter's lemma globally-generated-vanishing (killing H^1 against arbitrary module coefficients); Cohomology, lemma quasi-separated-cohomology-colimit (commutation of cohomology with direct sums on quasi-separated schemes); elementary homological algebra identifying an exact colimit-preserving endofunctor of Mod_A with tensoring by its value on A.

Source reference. Original lemma 01FF: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 21232–21232. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E7H: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 20971–20980. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Consider a commutative diagram*



Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

of morphisms of schemes. Assume $X \to S$ is flat, f is proper, $\dim(X_y) \leq 1$ *for $y \in Y$, and $R^1 f_* \mathcal{O}_X = 0$. Then $f_* \mathcal{O}_X$ is S-flat and formation of $f_* \mathcal{O}_X$ commutes with arbitrary base change $S' \to S$.*

Proof. We may assume Y and S are affine, say $S = \mathrm{Spec}(A)$. To show the quasi-coherent $\mathcal{O}_Y$-module $f_* \mathcal{O}_X$ is flat relative to S it suffices to show that $H^0(X, \mathcal{O}_X)$ is flat over A (some details omitted). By Lemma 0E7H we have $H^1(X, \mathcal{O}_X \otimes_A M) = 0$ for every A-module M. Since also $\mathcal{O}_X$ is flat over A we deduce the functor $M \mapsto H^0(X, \mathcal{O}_X \otimes_A M)$ is exact. Moreover, this functor commutes with direct sums by Cohomology, Lemma 01FF. Then it is an exercise to see that $H^0(X, \mathcal{O}_X \otimes_A M) = M \otimes_A H^0(X, \mathcal{O}_X)$ functorially in M and this gives the desired flatness. Finally, if $S' \to S$ is a morphism of affines given by the ring map $A \to A'$, then in the affine case just discussed we see that

$$H^0(X \times_S S', \mathcal{O}_{X \times_S S'}) = H^0(X, \mathcal{O}_X \otimes_A A') = H^0(X, \mathcal{O}_X) \otimes_A A'$$

This shows that formation of $f_* \mathcal{O}_X$ commutes with any base change $S' \to S$. Some details omitted. □